

Notes: Costain, Nuño & Thomas (JF, 2025)

The Term Structure of Interest Rates in a Heterogeneous Monetary Union

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Yield data and countries	2
2.2	Bond supply, fiscal forecasts, and PEPP scenarios	2
3	Models and empirical specifications	2
3.1	Yield decomposition	2
3.2	Expected default loss and credit risk premium	2
3.3	Fiscal pressure and endogenous default	3
3.4	PEPP channels	3
4	Identification and empirical design	3
5	Tables and figures with expanded interpretation	3
5.1	Figure 1: Effects of the pandemic and the PEPP announcement on German, Spanish, and Italian yields	3
5.2	Figure 2: Cross-sectional asset distributions + Figure 3: Baseline PEPP purchase scenario . .	4
5.3	Table I: Estimated Parameters	5
5.4	Figure 4: German and Italian yields, long-run behavior + Table II: Decomposing 10-year yields, spreads, and Sharpe ratios	5
5.5	Figure 5: IT-DE spreads, long-run behavior	6
5.6	Figure 6: Effects of PEPP announcement, baseline scenario ($\zeta = 1$) + Figure 7: Low-remittances scenario ($\zeta = 0$)	7
5.7	Table III: Robustness	7
5.8	Figure 8: Fiscal variables and default rate following pandemic and PEPP + Table IV: Event-conditional Sharpe ratios	8
5.9	Figure 9: Impact of the pandemic outbreak	9
6	Short summary	10
7	Expanded conclusion	10
7.1	Core conclusion	10
7.2	Economic intuition	10
7.3	Policy implication	11
7.4	Limitations	11
7.5	Takeaway	11

1 Big picture

- **Question.** How should we model yield curves in a monetary union where some sovereigns are default-free while others face default risk?
- **Empirical puzzle.** During the Covid-19 shock, Italian and Spanish yields moved much more than German yields, especially at the short end.
- **Model contribution.** The paper extends the Vayanos–Vila preferred-habitat term-structure framework to a heterogeneous monetary union with a default-free Core and a defaultable Periphery.
- **Yield decomposition.** Peripheral yields are decomposed into expected short rates, term premia, expected default losses, and credit risk premia.
- **New policy channel.** Central bank purchases reduce yields through a *default risk extraction* channel, not only through duration extraction.
- **Quantitative result.** Calibrating the model to Germany and Italy, both the level and the shifts in the sovereign spread are mainly attributable to the credit risk premium.

2 Data and measurement

2.1 Yield data and countries

The calibrated model maps Core to Germany and Periphery to Italy. The paper uses one-month, one-year, five-year, 10-year, and 20-year zero-coupon German and Italian sovereign yields, plus the one-month euro OIS rate for early periods.

Interpretation: The German curve disciplines default-free expected-rate and term-premium components. The Italian curve, relative to Germany, disciplines default-risk compensation.

2.2 Bond supply, fiscal forecasts, and PEPP scenarios

The paper constructs maturity distributions of ECB holdings and sovereign debt, plus fiscal forecasts and Eurosystem purchase paths. It compares pre-pandemic, pre-PEPP, and post-PEPP information sets.

Interpretation: The scenario design separates pandemic fiscal news from the PEPP asset-purchase announcement.

3 Models and empirical specifications

3.1 Yield decomposition

$$y_t(\tau) = y_t^{EX}(\tau) + y_t^{TP}(\tau) + y_t^{DL}(\tau) + y_t^{CR}(\tau), \quad y_t^*(\tau) = y_t^{EX*}(\tau) + y_t^{TP*}(\tau).$$

Interpretation: Expected rates and term premia apply to both countries. Expected default loss and credit risk premium apply only to the defaultable Periphery.

3.2 Expected default loss and credit risk premium

$$y_t^{DL}(\tau) = \frac{1}{\tau} E_t \int_0^\tau \delta \psi_{t+s} ds, \quad y_t^{CR}(\tau) = \frac{1}{\tau} E_t \int_0^\tau \xi_{t+s} ds.$$

Interpretation: The expected default loss compensates even a risk-neutral investor. The credit risk premium compensates risk-averse investors for bearing default risk.

3.3 Fiscal pressure and endogenous default

$$F_t = \int_0^\infty e^{-(\hat{r}+\phi)s} [d_{t+s} + f_{t+s}(0) - \zeta f_{t+s}^{CB}(0)] ds.$$

$$\psi_t = \eta \Phi \left(F_t + \frac{\bar{\Omega}}{\hat{r} + \phi} \right), \quad \psi_t = \underline{\psi} + \theta(F_t - \underline{F}) \text{ in the linearized calibration.}$$

Interpretation: If $\zeta > 0$, central bank holdings reduce fiscal pressure because redemptions on bonds held by the central bank can be rebated to the government.

3.4 PEPP channels

$$\text{PEPP} \Rightarrow \downarrow \text{net defaultable bond supply} \Rightarrow \downarrow y^{CR}$$

$$\text{PEPP with } \zeta > 0 \Rightarrow \downarrow F_t \Rightarrow \downarrow \psi_t \Rightarrow \downarrow y^{DL}, \downarrow y^{CR}.$$

4 Identification and empirical design

The German curve identifies the safe term-structure block. The Italian-German spread identifies default compensation because monetary policy is common across the union. The PEPP announcement is used as a policy shock to net bond supply and fiscal-pressure expectations. The main limitation is that affine tractability requires deterministic fiscal and purchase paths.

5 Tables and figures with expanded interpretation

5.1 Figure 1: Effects of the pandemic and the PEPP announcement on German, Spanish, and Italian yields

Read: Peripheral curves moved much more than the German curve, including at the short end. This pattern motivates adding default risk to the preferred-habitat term-structure model.

Detailed interpretation: A duration-extraction channel should primarily move long maturities and should not create such asymmetric short-end shifts across euro-area sovereigns. The figure therefore motivates the default-risk channel.

Economic message: In the euro area, the same monetary policy shock has different effects across sovereign issuers because default risk differs.

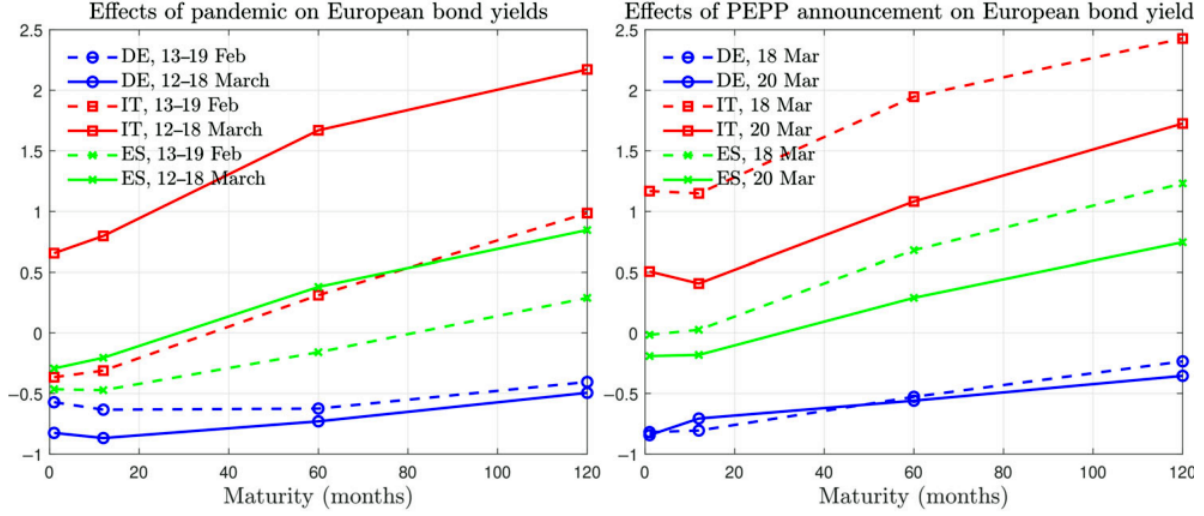


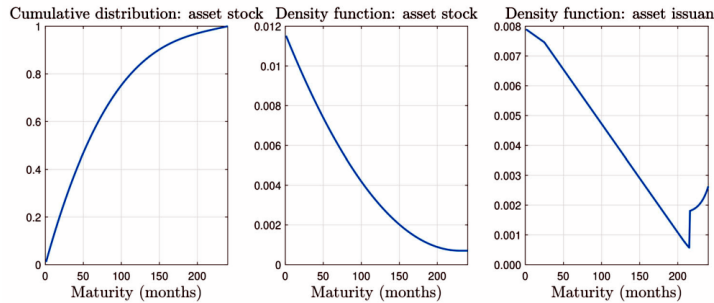
Figure 1: Effects of the pandemic and PEPP announcement on yields

5.2 Figure 2: Cross-sectional asset distributions + Figure 3: Baseline PEPP purchase scenario

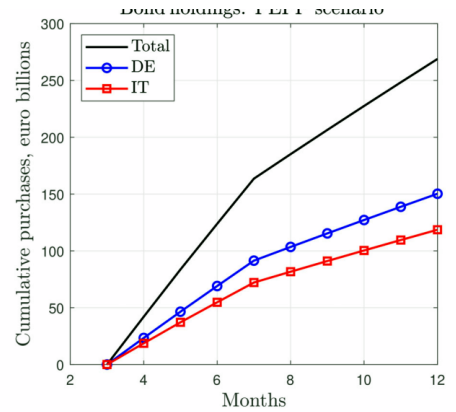
Read: Figure 2 shows the maturity distributions used for asset stocks and issuance. Figure 3 shows the baseline PEPP purchase path for German and Italian bonds.

Detailed interpretation: The maturity distribution determines where risk is removed from private portfolios. The purchase path determines when the removal happens. These are key inputs into duration and default risk extraction.

Economic message: The total PEPP envelope matters, but its maturity and timing composition also matter.



(a) Figure 2: Cross-sectional asset distributions



(b) Figure 3: Baseline PEPP purchase scenario

5.3 Table I: Estimated Parameters

Read: Table I reports risk aversion, default-rate intercept and slope, PH demand parameters, and the remittance coefficient.

Detailed interpretation: The most important parameters are risk aversion γ , the default-rate level ψ , the default-rate slope θ , and the remittance coefficient ζ . Together they determine the scale of credit risk premia and the strength of the endogenous-default PEPP channel.

Economic message: The calibration makes default-risk premia large without requiring implausible default probabilities.

Table I Estimated Parameters	
Parameters	Values
γ : Risk aversion	0.0108
ψ : Default rate intercept	86.2 bps
θ : Slope of default rate	2.05 bps/€billion
$\bar{r} + \phi$: Discount rate in fiscal pressure	30.7%
κ_h : Mean-reversion of PH shocks	0.00535
σ_h : Volatility of PH demand innovations	€499 billion
ν_h : Relative volatility of IT PH demand	0.300
χ_h : Demand correlation parameter	-0.558
α_h : Slope parameter of PH demand	€80.0 billion/pp
ζ : Remittance rule coefficient	1

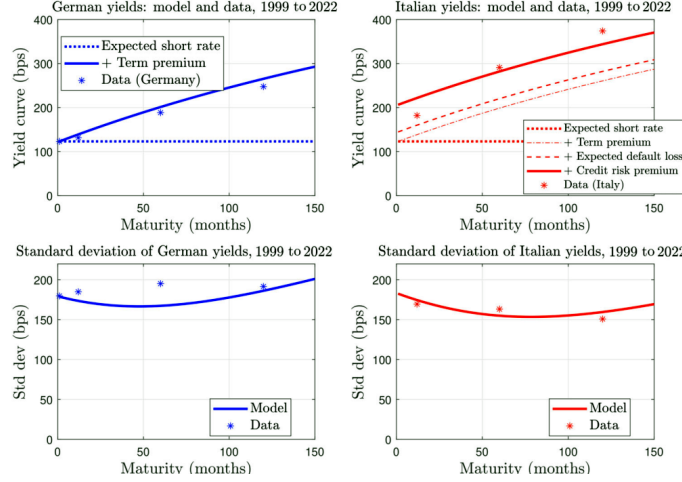
Table I: Estimated parameters

5.4 Figure 4: German and Italian yields, long-run behavior + Table II: Decomposing 10-year yields, spreads, and Sharpe ratios

Read: Figure 4 decomposes German and Italian yield curves. Table II reports the 10-year yield/spread decomposition and Sharpe ratios. The model spread is 0.79%, with expected default loss 0.22% and credit risk premium 0.62%.

Detailed interpretation: The table shows that credit risk premium dominates expected default loss. This is the central long-run quantitative finding.

Economic message: The Italian-German spread is primarily compensation for bearing default risk, not merely actuarial expected loss.



(a) Figure 4: Long-run yields

Table II
Decomposing 10-Year Yields, Spreads, and Sharpe Ratios

This table reports the instantaneous Sharpe ratios $S_t^Y(\tau)$ and $S_t^S(\tau)$, interpreting the time unit as one year, for $\tau = 10$ years, and related yield components. $\dagger$ indicates quantities stated in annualized percentage points. Model notation is used to clarify the quantities shown, but the function argument (τ) is omitted for brevity.

	DE 10Y Yield		IT-DE 10Y Spread	
	Variable	Value	Variable	Value
Yield/Spread: data $\dagger$	—	2.47%	—	1.26%
Yield/Spread: model $\dagger$	$E_{Y_t}^{\text{model}}$	2.65%	$E_{Y_t} - E_{Y_t}^{\text{model}}$	0.79%
Expected rates/default loss $\dagger$	$E_{Y_t}^{\text{model}} - \bar{r}$	1.23%	$E_{Y_t}^{\text{model}} - \bar{r}_D$	0.22%
Term/credit premiums $\dagger$	$E_{Y_t}^{\text{model}} - \bar{r}_D$	1.42%	$E_{Y_t}^{\text{model}} - \bar{r}_D$	0.62%
Sharpe ratio: data 1999–2022	—	0.51	—	0.23
Sharpe ratio: model	$S_t^Y(\tau)$	0.57	$S_t^S(\tau)$	0.27
Numerator $\dagger$	$A^{\top} E_{\Delta_t}$	2.70%	E_{Δ_t}	0.62%
Denominator $\dagger$	$\sqrt{A^{\top} \Sigma \Sigma^{\top} A}$	4.72%	$\delta \sqrt{V}$	2.32%

(b) Table II: Yield/spread decomposition

5.5 Figure 5: IT-DE spreads, long-run behavior

Read: Figure 5 isolates the Italian-German spread and shows that the credit risk premium is the dominant model component.

Detailed interpretation: The model captures the importance of default-risk compensation but produces a flatter spread curve than the data, pointing to the limitation of a constant long-run default rate.

Economic message: The main result is about components: the spread is mostly risk premium, even if the simple model misses some slope.

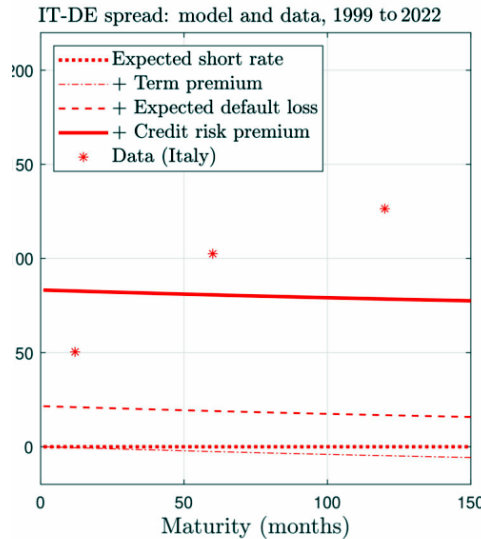


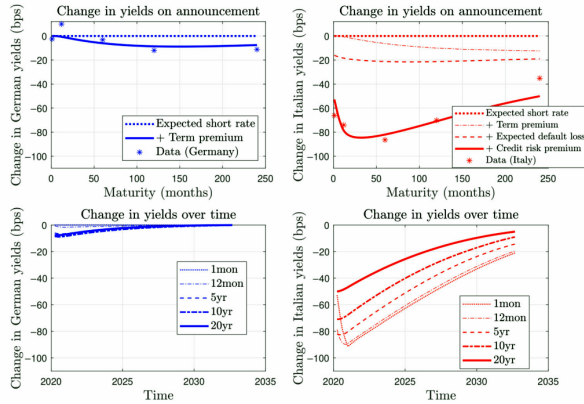
Figure 5: IT-DE spreads, long-run behavior

5.6 Figure 6: Effects of PEPP announcement, baseline scenario ($\zeta = 1$) + Figure 7: Low-remittances scenario ($\zeta = 0$)

Read: Figure 6 shows that PEPP lowers Italian yields much more than German yields, mainly through the credit risk premium. Figure 7 removes the remittance/default-probability channel.

Detailed interpretation: The comparison shows that quantity-based default risk extraction remains important even when $\zeta = 0$, but the remittance channel improves the short-end fit and amplifies the fall in Italian yields.

Economic message: PEPP's peripheral impact is not a pure duration effect; it is mostly a default-risk effect.



(a) Figure 6: PEPP baseline

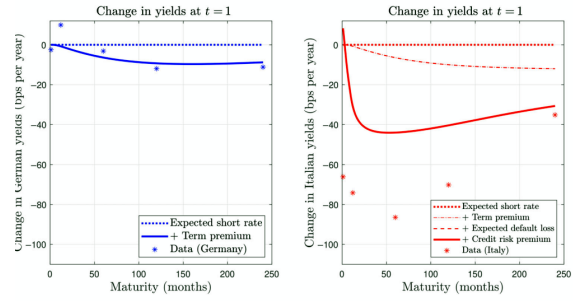


Figure 7. Effects of PEPP announcement: Low-remittances scenario ($\zeta = 0$). Stars: Shift

(b) Figure 7: Low-remittances scenario

5.7 Table III: Robustness

Read: The benchmark credit risk premium contribution is 0.74 in the long run and 0.81 for PEPP. The result is robust across many perturbations, but weakens under very low risk aversion or very low haircuts.

Detailed interpretation: The robustness table shows that the central result is not a knife-edge artifact. The cases that make expected default losses more important also worsen fit or require implausibly low risk compensation.

Economic message: Credit risk premia dominate under parameter values consistent with the observed term structure.

Table III
Robustness

This table compares model-generated statistics across different calibrations. The benchmark calibration is described in Table I. The following rows report the effect of changing one parameter by the stated amount, relative to its benchmark value. Model fit refers to the sum of squared deviations of the statistics, expressed in percentage points, listed in the bullet points in Section III.C. Credit risk premium contribution refers to the credit risk premium as a fraction of the total compensation for default risk at 10-year maturity, either in the long run or (in differences) in the effects of the PEPP announcement. * indicates annual yields on 10-year bonds, expressed in basis points.

Calibration	Model Fit	Credit Risk Prem. Contribution		Mean Yields*		PEPP Impact*
		Long Run	PEPP	DE	IT	
Data	—	—	—	247	373	−71
Benchmark	0.548	0.74	0.81	265	343	−71
Parameters calibrated directly from observables						
$\kappa \times 1.25$	1.063	0.74	0.81	252	331	−70
$\sigma \times 1.25$	3.306	0.74	0.81	336	411	−73
$\delta = 0.1$	2.854	0.54	0.68	268	286	−24
$\delta = 0.05$	3.817	0.37	0.52	268	275	−16
Parameters estimated by minimizing the distance criterion						
$\gamma \div 3$	5.448	0.50	0.63	170	213	−35
$\gamma \div 6$	8.356	0.33	0.47	145	177	−23
$\psi \times 1.25$	0.663	0.74	0.82	264	362	−71
$\bar{\theta} \times 1.25$	0.633	0.74	0.81	265	343	−84
$(\bar{\rho} + \phi) \times 1.25$	0.593	0.74	0.82	265	343	−62
$\sigma_h \times 1.25$	0.724	0.74	0.81	268	345	−73
$\alpha_h \times 1.5$	0.653	0.73	0.80	253	328	−64
$\zeta = 0$	1.516	0.74	1.00	265	343	−40

Table III: Robustness

5.8 Figure 8: Fiscal variables and default rate following pandemic and PEPP + Table IV: Event-conditional Sharpe ratios

Read: Figure 8 shows fiscal pressure, default probability, expected default loss, default risk price, and credit risk premium rising after the pandemic and falling after PEPP. Table IV shows the Italian spread Sharpe ratio rising from 0.362 to 0.521 and then falling to 0.489.

Detailed interpretation: Figure 8 opens the black box of the mechanism. Table IV shows that the implied risk compensation is economically meaningful but not implausibly large.

Economic message: The default-risk channel operates through a coherent fiscal-pressure mechanism.

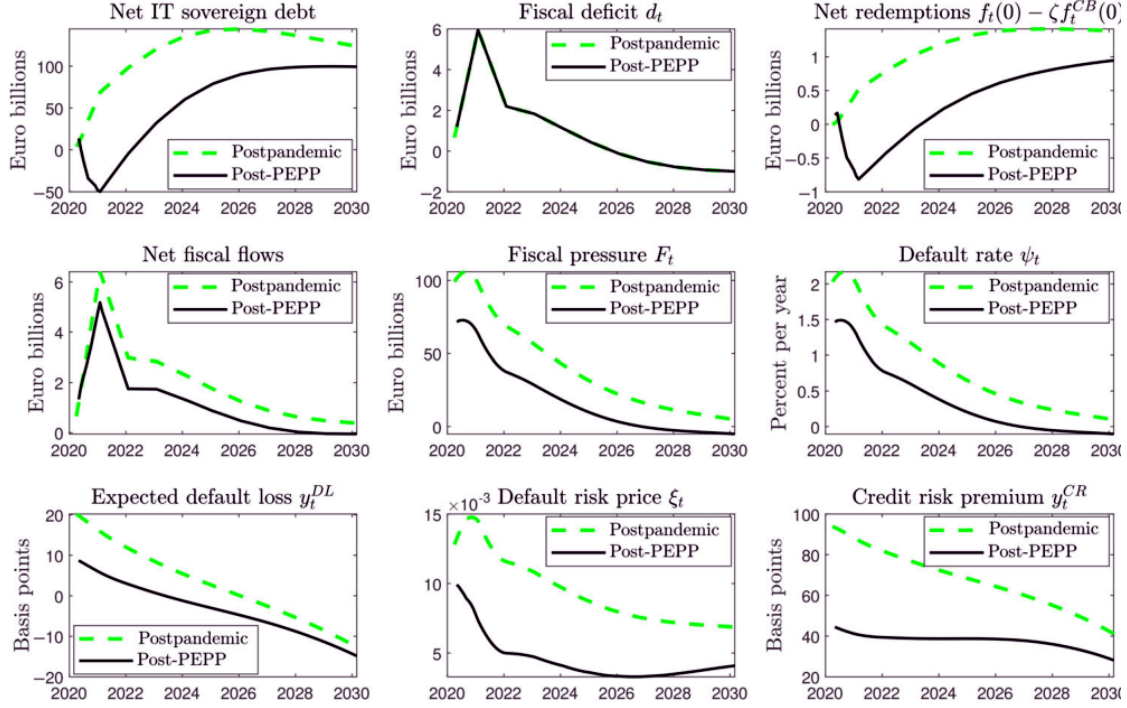


Figure 8: Fiscal variables/default rate

Table IV
Model-Implied Sharpe Ratios, Conditional on Events of February to March 2020

This table reports the model-implied instantaneous Sharpe ratios $S_t^*(\tau)$ and $S_t^{\Delta}(\tau)$, interpreting the time unit as one year, for $t = \text{March 2020}$ and $\tau = 10$ years, conditional on the prepandemic, pre-PEPP, and post-PEPP information sets. † indicates quantities stated in annualized percentage points. Model notation is used to clarify the quantities shown.

	DE 10Y Yield	IT-DE 10Y Spread
Sharpe ratio		
Prepandemic	0.399	0.362
Pre-PEPP	0.394	0.521
Post-PEPP	0.407	0.489
Sharpe numerator†	$A_t^*(\tau)^\top \lambda_t$	δ_t
Prepandemic	1.93%	1.22%
Pre-PEPP	1.90%	2.58%
Post-PEPP	1.96%	2.21%
Sharpe denominator†	$\sqrt{A_t^*(\tau)^\top \Sigma \Sigma^\top A_t^*(\tau)}$	$\delta \sqrt{\psi_t}$
Prepandemic	4.82%	3.37%
Pre-PEPP	4.83%	4.96%
Post-PEPP	4.83%	4.53%

Table IV: Event-conditional Sharpe ratios

5.9 Figure 9: Impact of the pandemic outbreak

Read: The model generates a large upward shift in Italian yields after the pandemic shock and a smaller movement in German yields. For Italian 10-year bonds, the paper reports roughly 20 bps from expected default loss and roughly 100 bps from the credit risk premium.

Detailed interpretation: Figure 9 is an out-of-sample test of the same mechanism in the opposite direction from PEPP: fiscal stress raises default-risk compensation.

Economic message: The same model explains both the pandemic stress episode and PEPP stabilization.

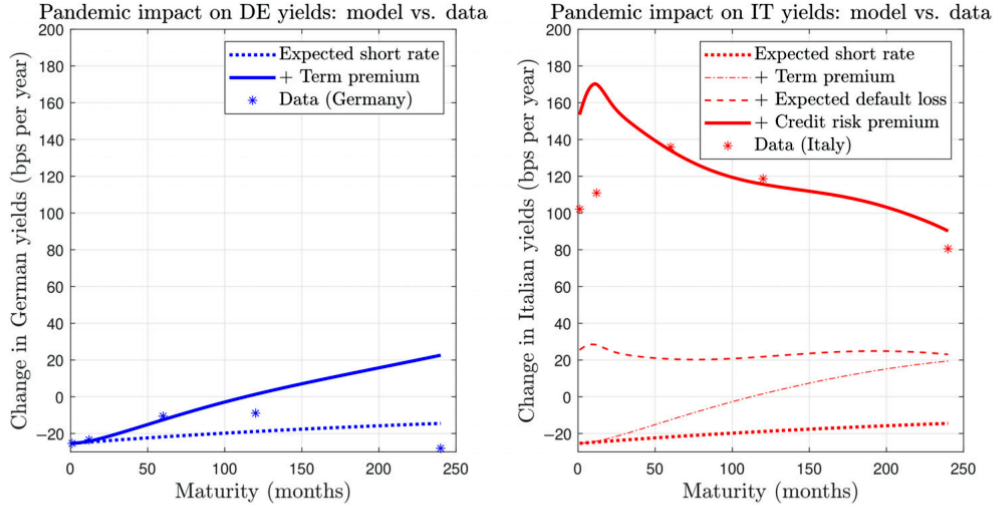


Figure 9. Impact of the pandemic outbreak: German and Italian yields, February to March 2020. Stars: Shift in weekly average German yields (left) and Italian yields (right) from week of February 13 to 19, 2020 to week of March 12 to 18, 2020. Lines: Model-generated change

Figure 9: Pandemic outbreak effects

6 Short summary

- The paper models a monetary union with a default-free Core and a defaultable Periphery.
- Peripheral yields contain expected rates, term premia, expected default losses, and credit risk premia.
- The Italy-Germany spread is mainly a credit risk premium.
- PEPP lowers Italian yields primarily through default risk extraction.
- Endogenous default makes purchases more powerful because they reduce fiscal pressure and default probability.
- Pandemic fiscal news works in the opposite direction by raising default probability and the default risk price.

7 Expanded conclusion

7.1 Core conclusion

Costain, Nuño, and Thomas show that euro-area sovereign yield curves cannot be understood using a safe-asset duration-risk model alone. The Italian-German spread is mostly a credit risk premium, and PEPP works mainly by reducing that premium.

7.2 Economic intuition

Risk-averse arbitrageurs must hold residual sovereign risk. When the residual supply is defaultable, they demand default-risk compensation. Central bank purchases reduce the amount of defaultable risk in private hands and, under the remittance channel, reduce future fiscal pressure.

7.3 Policy implication

Asset purchases in a heterogeneous monetary union are not merely duration policy. They are also sovereign-risk stabilization policy.

7.4 Limitations

The affine solution requires deterministic fiscal and purchase paths. The long-run model also produces a flatter sovereign spread curve than observed in the data.

7.5 Takeaway

For peripheral sovereigns, default risk extraction is likely the dominant channel of asset-purchase transmission.